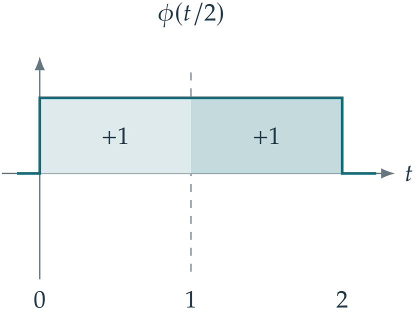
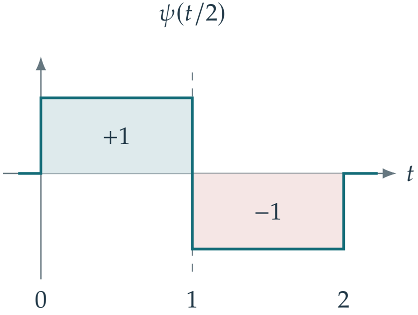


$\langle \phi(\cdot/2), \phi(\cdot - n)\rangle = 1$ for $n = 0, 1$

$$h_0 = h_1 = 1/\sqrt{2}$$



$\langle \psi(\cdot/2), \phi(\cdot - n)\rangle = (-1)^n$ for $n = 0, 1$

$$g_0 = 1/\sqrt{2}, \quad g_1 = -1/\sqrt{2}$$

$\phi = \mathbf{1}_{[0,1)}, \quad \psi = \mathbf{1}_{[0,1/2)} - \mathbf{1}_{[1/2,1)}; \qquad h_n = g_n = 0$ for $n \notin \{0, 1\}$.